4 Theorem If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

where $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$

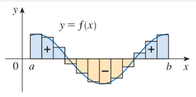


FIGURE 3
$\sum f(x_i^*)\,\Delta x$ is an approximation to the net area.

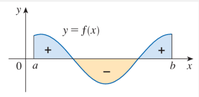


FIGURE 4
$\int_a^b f(x)\,dx$ is the net area.

EXAMPLE 2 Express

$$\lim_{n\to\infty} \sum_{i=1}^{n} (x_i^3 + x_i \sin x_i)\,\Delta x$$

as an integral on the interval $[0, \pi]$.

$$\int_0^{\pi} (x^3 + x\sin x)\,dx$$

Sums of Powers

5 $\displaystyle\sum_{i=1}^{n} 1 = n$

6 $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

7 $\displaystyle\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

8 $\displaystyle\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$

Properties of Sums

9 $\displaystyle\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i$

10 $\displaystyle\sum_{i=1}^{n} (a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$

11 $\displaystyle\sum_{i=1}^{n} (a_i - b_i) = \sum_{i=1}^{n} a_i - \sum_{i=1}^{n} b_i$

4 Theorem If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

where $\Delta x = \dfrac{b-a}{n}$ and $x_i = a + i\,\Delta x$

EXAMPLE 3 Evaluate $\int_0^3 (x^3 - 6x)\,dx$.

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} \left(\left(\frac{3i}{n}\right)^3 - \left(\frac{18i}{n}\right)\right)\cdot\frac{3}{n} = \lim_{n\to\infty} \frac{3}{n}\sum_{i=1}^{n}\left(\frac{27i^3}{n^3} - \frac{18i}{n}\right) = \lim_{n\to\infty} \frac{3}{n}\left[\left(\frac{27}{n^3}\right)\sum_{i=1}^{n} i^3 - \left(\frac{18}{n}\right)\sum_{i=1}^{n} i\right] = \lim_{n\to\infty}$$

$$\Delta x = \frac{3-0}{n}$$

$$\frac{81}{n}\left[\frac{n(n+1)}{2}\right]^2 \qquad \frac{54}{n^2}\left(\frac{n(n+1)}{2}\right)$$

$$\frac{81}{4}\left(\frac{n^2+n}{n^2}\right)^2 \qquad 27\left(\frac{n^2+n}{n^2}\right)$$

$$\frac{81}{4}\left(1+\frac{1}{n}\right)^2 \qquad 27\left(1+\frac{1}{n}\right)$$

$$= \lim_{n\to\infty}\left[\frac{81}{4}\left(1+\frac{1}{n}\right)^2 - 27\left(1+\frac{1}{n}\right)\right]$$

$$\frac{81}{4} - 27$$

$$20.25 - 27 = -6.75$$

1) Found delta x
2) Found xi
3) We plugged in xi into the function
4) Solved the Riemann sums using the properties

EXAMPLE 4
(a) Set up an expression for $\int_1^3 e^x\,dx$ as a limit of sums.

$$\Delta x = \frac{3-1}{n} = \frac{2}{n}$$

$$x_i = 1 + i\left(\frac{2}{n}\right) = 1 + \frac{2i}{n}$$

$$\lim_{n\to\infty} \sum_{i=1}^{n=\infty} e^{1+\frac{2i}{n}}\left(\frac{2}{n}\right) = \lim_{n\to\infty} \frac{2}{n}\sum_{i=1}^{n=\infty} e^{1+2i/n}$$

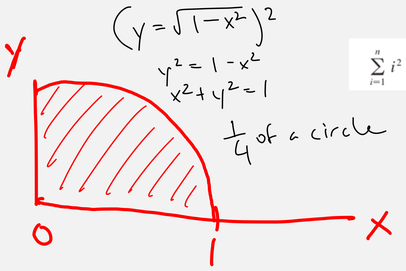
EXAMPLE 5 Evaluate the following integrals by interpreting each in terms of areas.
(a) $\int_0^1 \sqrt{1-x^2}\,dx$ (b) $\int_0^3 (x-1)\,dx$

$$\left(y = \sqrt{1-x^2}\right)^2$$

$$y^2 = 1 - x^2$$

$$x^2 + y^2 = 1$$

$\frac{1}{4}$ of a circle



$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

x	y
0	
1	

Homework do Problems from 5.2 as many as you can